VIORA-IOT Based Smart Assistive Device for Visually Impaired People Using Obstacle Detection And Voice Feedback

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Abstract—**VIORA is an IoT-based smart assistive device developed to improve the safety and mobility of visually impaired individuals through real-time obstacle detection and feedback. The system utilizes an ESP32 microcontroller, ultrasonic sensors, and a DFPlayer Mini module to detect obstacles in multiple directions and provide instant voice alerts. Designed as a wearable solution such as a belt or cap, it ensures ease of use, low cost, and reliability. By integrating simple hardware components with efficient processing, the system offers an effective and practical alternative to complex assistive technologies, helping users navigate their surroundings independently and safely.**

Index Terms— **Visually Impaired, IoT, Obstacle Detection, Ultrasonic Sensor, ESP32, Assistive Technology, Voice Feedback.**

I. Introduction

Navigating through unfamiliar or crowded environments presents a significant challenge for visually impaired individuals, as they are unable to perceive obstacles such as walls, poles, vehicles, or pedestrians in real time. This limitation often leads to accidents, anxiety, and a strong dependence on others for mobility. Traditional navigation aids like the white cane, although widely used, have several limitations. They can only detect obstacles at ground level and within a short range directly in front of the user, offering no information about objects at head or chest level or from the sides. As a result, users remain vulnerable to unexpected collisions with objects like tree branches, signboards, or moving obstacles.

With the rapid development of the Internet of Things (IoT) and embedded systems, new possibilities have emerged for creating advanced assistive technologies that can address these challenges. By integrating sensors, microcontrollers, and real-time feedback systems, modern devices can provide a more complete understanding of the user's surroundings. VIORA is a smart, wearable assistive device developed as an electronic travel aid to enhance the mobility and safety of visually impaired individuals. The system consists of a compact hardware unit built around an ESP32 microcontroller and three HC-SR04 ultrasonic sensors strategically positioned to monitor obstacles in the left, front, and right directions.

When an obstacle is detected within a predefined distance, the device instantly provides clear and directional voice guidance through an earphone or speaker. This real-time audio feedback allows the user to easily understand the position of obstacles and take appropriate action, effectively creating an "auditory map" of their surroundings. Unlike traditional systems that rely on vibrations or unclear sound signals, VIORA focuses on intuitive voice alerts, reducing the learning curve and making it more user-friendly.

The development of VIORA is rooted in the long-standing need to improve independent mobility for visually impaired individuals. While existing solutions like guide dogs and advanced electronic aids are available, they often involve high costs, complex usage, or limited accessibility. VIORA aims to overcome these barriers by offering a solution that is affordable, simple to operate, durable, and suitable for everyday use. It addresses key issues such as limited perception, lack of directional awareness, complex feedback mechanisms, and poor wearability found in many existing devices.

The primary motivation behind this project is to empower visually impaired individuals by increasing their independence, confidence, and quality of life. By providing continuous, real-time environmental information, VIORA reduces the fear of accidents and allows users to navigate more freely and safely. The system is designed to be lightweight, power-efficient, and reliable, ensuring consistent performance in different conditions.

Overall, VIORA represents an effort to combine modern IoT technology with practical design to create an effective assistive device. It aims to bridge the gap between traditional mobility aids and advanced smart systems, contributing towards a more inclusive society where visually impaired individuals can move independently with greater safety and confidence.

II. LITERATURE SURVEY

Research in assistive technology for visually impaired individuals has introduced multiple sensor-based and AI-driven solutions [5], [9]. Projects like the **Smart Blind Stick** (Balakrishnan et al., 2024) use Arduino Mega with static and rotating ultrasonic sensors [2], improving side detection but relying on buzzer/vibration feedback, which is not intuitive. Advanced systems such as **Smart Glass for Visually Impaired People** (Kavya et al., 2024) use Raspberry Pi, YOLO, and OCR for object and text detection [3], but these require high processing power, are costly, and more complex to operate. Low-cost wearable solutions like those developed by Swami et al. [4] using STM32 with a single ultrasonic sensor are compact and affordable but lack multi-directional detection and provide only buzzer alerts. Overall, existing works show a trade-off between cost, complexity, and feedback clarity [10], highlighting the need for a balanced solution.

Comparative analysis shows that traditional aids like the White Cane [1], [9] provide basic ground-level obstacle sensing but lack upper-body protection and directional awareness. Smart Sticks [2] improve detection but are limited to forward sensing and require manual handling. Advanced IoT wearables using cameras and processors [3], [5] offer more features but are expensive and power-intensive. In contrast, the VIORA Smart Belt uses a multi-directional ultrasonic sensor setup [6] and an ESP32 controller [8] to offer wide detection coverage up to 200 cm, giving 180° awareness (Left, Front, Right). Its voice-based feedback system [7] is clearer and more intuitive than vibration-only devices [10], and its low-cost design supports accessibility emphasized by Swami et al. [4]. VIORA also provides a hands-free form factor with potential for future GPS and AI integration [5], making it more practical than handheld solutions.

From the review and comparison, clear limitations of existing systems emerge. Many provide non-intuitive feedback through buzzers or vibrations, requiring users to memorize patterns. Devices with limited field of view cannot specify obstacle direction, giving only a general alert. High-end camera-based systems are costly, complex, and power-hungry. Some devices are bulky or uncomfortable for daily use, affecting wearability. Sensor performance may drop in rain, sunlight, or humidity. Many systems lack user-adjustable detection ranges, reducing adaptability in different environments. These limitations emphasize the need for a robust, intuitive, and multi-directional assistive device suitable for real-world use, which VIORA aims to fulfill [2], [3], [5].

III. SYSTEM ANALYSIS & DESIGN A.System Analysis

This section provides an overview of the system architecture for the VIORA assistive device, including its requirements, feasibility, and design approach. The system is built around the ESP32 microcontroller, three HC-SR04 ultrasonic sensors, and a

DFPlayer Mini audio module, all integrated to deliver real-time obstacle detection and directional voice feedback. The design process covers functional and non-functional requirements, hardware and software specifications, and a feasibility assessment to ensure the solution is practical, affordable, and technically robust.

The functional requirements include continuous obstacle detection in three directions (left, front, right), accurate distance measurement up to 200 cm, and instant voice alerts when an obstacle falls within a predefined range. The device must also handle power management through a rechargeable battery, initialize all components on startup, and filter out false echoes to avoid incorrect alerts. Non-functional requirements focus on performance (response time < 500 ms), reliability in indoor/outdoor conditions, usability through a single-button operation, portability as a wearable device, power efficiency for 8–10 hours of use, and electrical safety through low-voltage operation and comfortable audio levels.

The hardware setup consists of the ESP32 board, three HC-SR04 sensors, a DFPlayer Mini with a 3W speaker, a 5V Li-ion/Li-Po battery with an LM7805 regulator, and a compact, lightweight enclosure suitable for a belt or cap. Supporting components such as wiring, power switches, and optional LEDs complete the build. The software requirements include Windows/Linux for development, Arduino IDE or PlatformIO, C/C++ firmware, and libraries such as ESP32 board package, NewPing, and DFPlayerMini. Audio prompts are recorded using tools like Audacity.

The system is technically feasible because all components are lightweight, low-cost, and widely supported with existing libraries. The ESP32 is capable of handling three sensor inputs and audio operations simultaneously, while the DFPlayer Mini simplifies voice playback. The modular architecture ensures easy integration and future scalability, such as Bluetooth-based features. Economically, the project is highly cost-effective, with total component expenses well under $50 USD. Open-source software further reduces costs, and the use of replaceable, mass-produced parts ensures long-term affordability and sustainability. Overall, the VIORA design is practical, reliable, and well-suited for real-world deployment.

B.Project Design

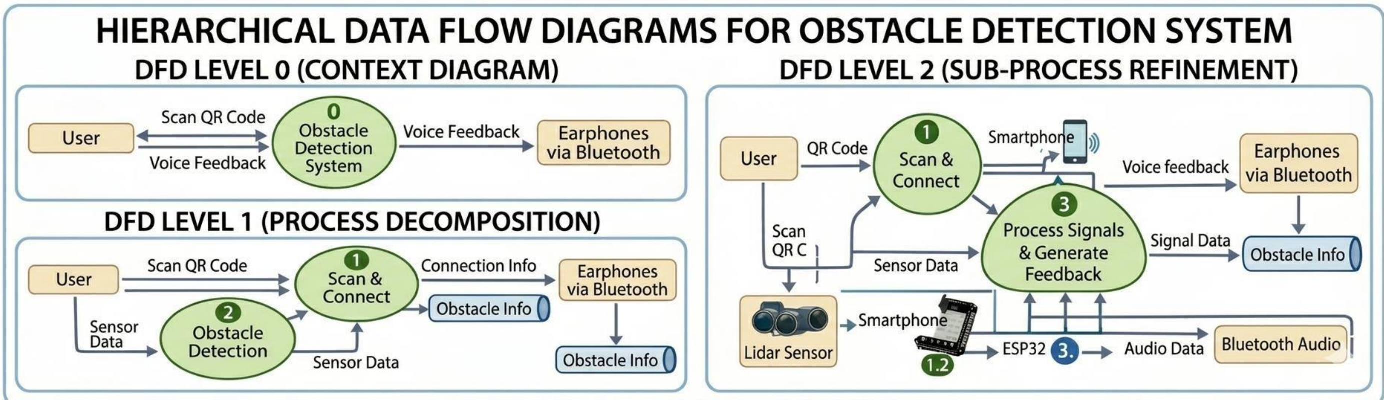

Fig: 1 DFD Diagram

The proposed system is a smart obstacle detection and voice feedback system designed to assist visually impaired users in navigating their surroundings safely. It integrates QR code technology, sensors, a microcontroller (ESP32), and Bluetooth communication to provide real-time guidance. The system begins with the user scanning a QR code using a smartphone, which initializes the device and establishes a Bluetooth connection between the smartphone, ESP32 module, and wireless earphones. This step ensures that all components are properly connected and ready for operation.Once connected, the obstacle detection phase begins. Sensors such as ultrasonic sensors continuously monitor the environment and collect data about nearby objects. These sensors measure the distance and detect obstacles in different directions like front, left, and right. The collected sensor data is then transmitted to the ESP32 microcontroller, which acts as the central processing unit of the system.

C.Flowchart

Fig. 2: VIORA Smart Belt System - Operational Flowchart and Functional Architecture

IV. Methodology & Implementation

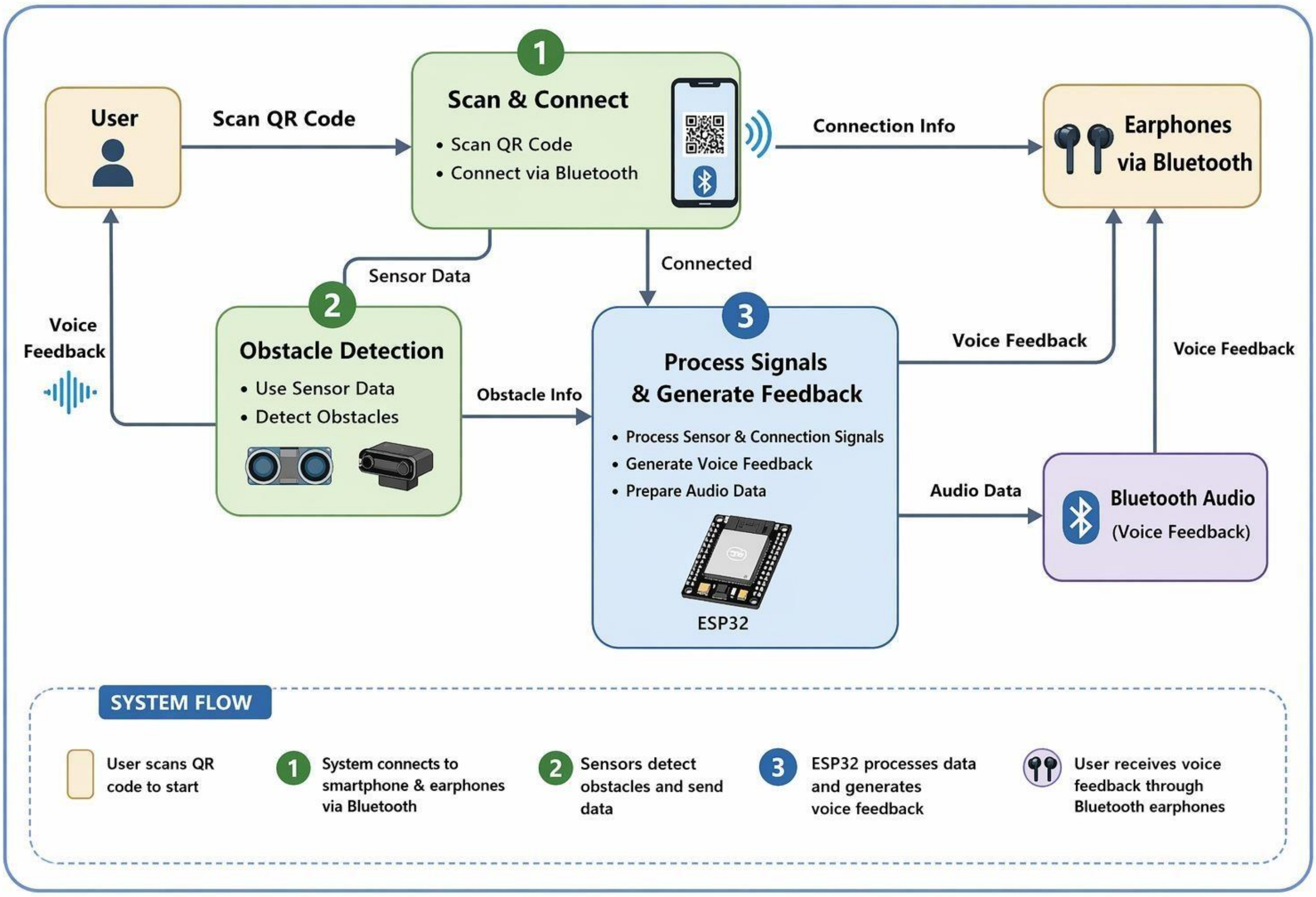

Fig : 3 Methodology Diagram

The VIORA system is developed using an integrated approach that combines hardware architecture, software logic, and useroriented design. The block diagram represents the complete flow of signals between components, showing how the ESP32 microcontroller communicates with the ultrasonic sensors, DFPlayer Mini, speaker, and power supply to generate real-time audio alerts for the user. Figure 4.A illustrates the Modular Layered Architecture of VIORA – IoT-Based Smart Assistive Belt.

The project uses an ESP32 development board as the main controller because of its dual-core processing, built-in Wi-Fi/Bluetooth features, and low power consumption, making it suitable for real-time operations and future IoT connectivity. HC-SR04 ultrasonic sensors are used for obstacle detection, offering a distance range of 2 cm to 400 cm with high accuracy, while the DFPlayer Mini module provides audio output through stored MP3 files on a microSD card. Embedded C/C++ with the Arduino framework is used for programming, and the Arduino IDE supports code development and firmware uploading. Libraries such as NewPing for ultrasonic handling and DFRobotDFPlayerMini for audio control simplify implementation. A 5V rechargeable battery with an LM7805 voltage regulator ensures stable power delivery, with capacitors added to reduce electrical noise.

The hardware implementation includes integrating the ESP32, three ultrasonic sensors, DFPlayer Mini, and a speaker onto a PCB or prototyping board for durability and compact size. The left, front, and right sensors are mounted at approximately 45-degree angles on a wearable belt or cap to cover the user's forward and side detection zones. A regulated power circuit supplies clean 5V to all components, and a 3D-printed enclosure protects the electronics while providing slots for the sensors and speaker.

The software implementation begins with initializing serial communication, configuring GPIO pins, and setting the DFPlayer Mini volume inside the setup() function. The loop() function continuously reads distances by triggering each ultrasonic sensor and calculating the echo response time. These distance values are compared with a fixed threshold (e.g., 100 cm) to detect nearby obstacles. When a sensor detects an object within the threshold and the cooldown period has passed, the system plays the corresponding direction-specific audio prompt using the playSound() function. A cooldown timer is implemented to avoid repetitive alerts for the same obstacle, ensuring that feedback remains clear and non-disruptive for the user

V. Results & Evaluation

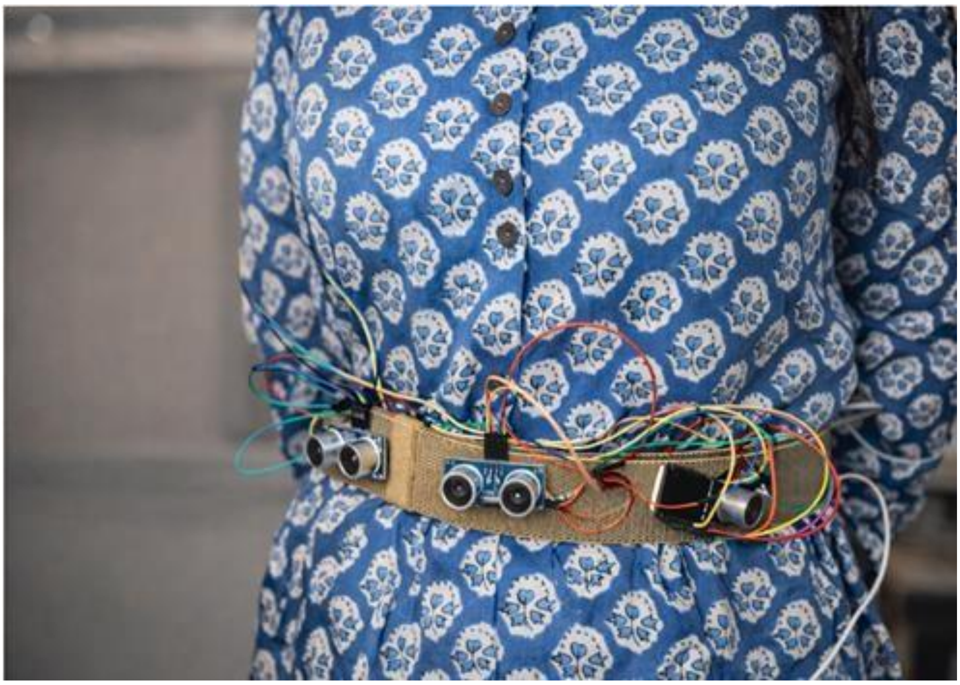

Fig : 4

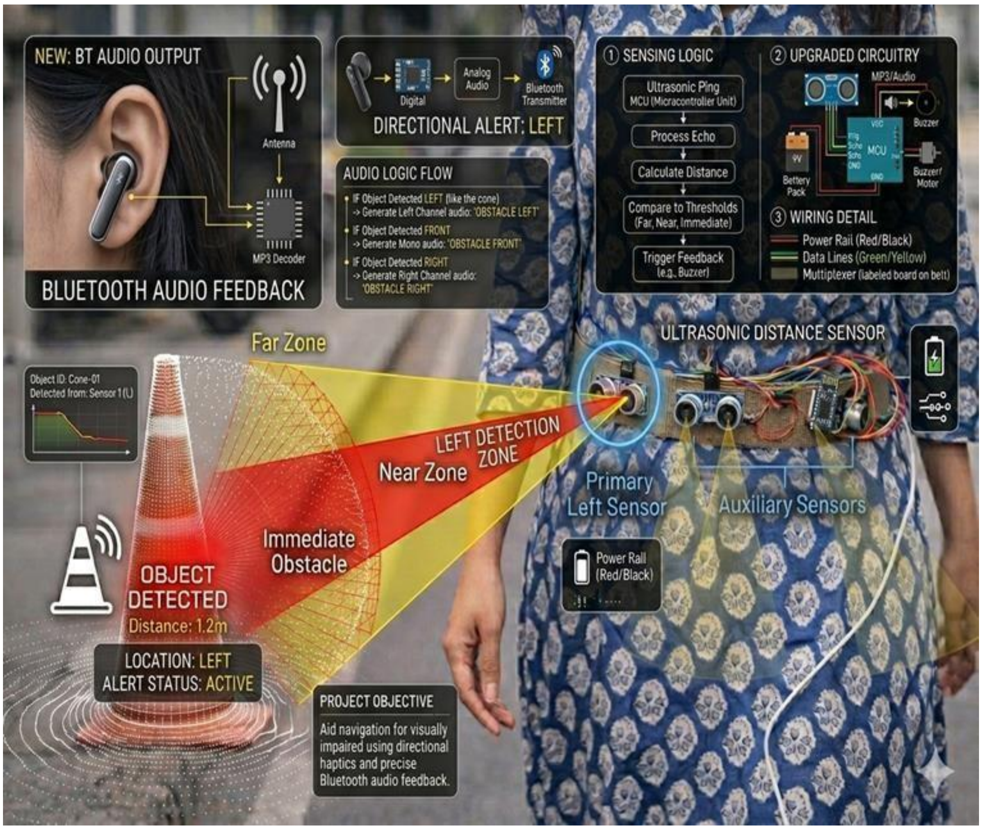

Fig : 5

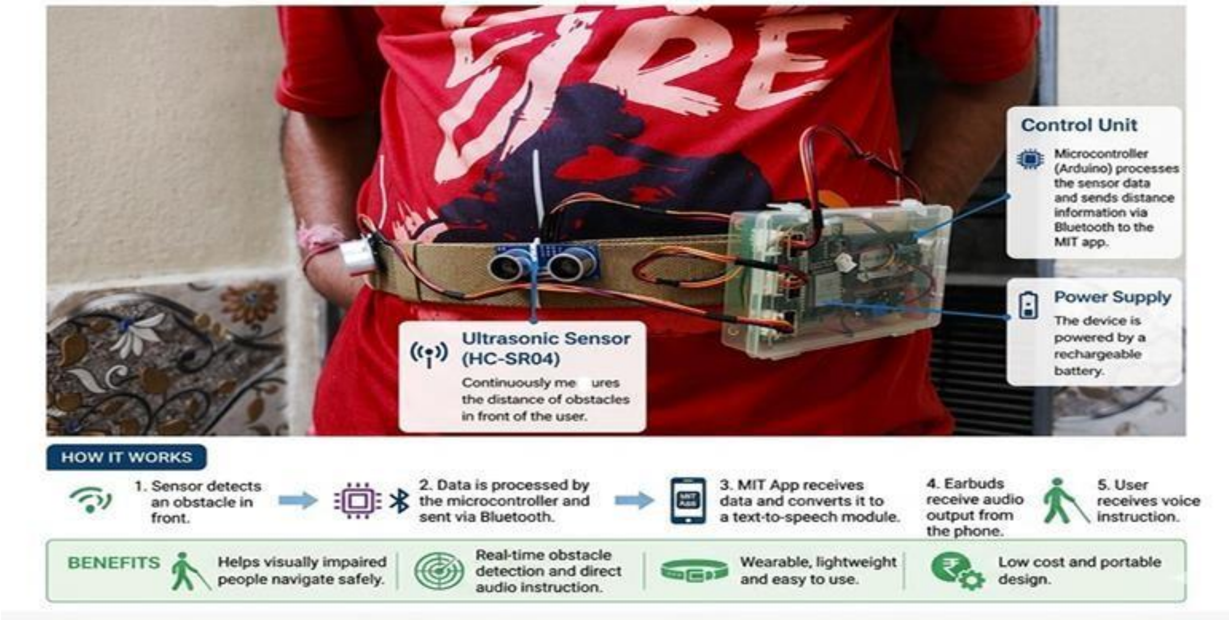

Fig : 6

The VIORA prototype was subjected to a series of tests to evaluate its performance in terms of obstacle detection accuracy, response time, and user feedback clarity. Testing was conducted in various indoor and outdoor environments with different lighting conditions. The system consistently demonstrated its ability to detect obstacles placed within its defined range and provide the appropriate directional voice alert.

The response time from obstacle placement to audio alert was measured to be consistently under 500 milliseconds, which is well within the target specification and fast enough for a user to react in real-time. The detection range of the ultrasonic sensors was reliable from 10cm up to 200cm. The sensors showed minimal false positives, though accuracy decreased slightly with soft or sound-absorbing materials like fabric, which is a known limitation of ultrasonic technology. The voice prompts generated by the DFPlayer Mini were clear and loud enough to be easily heard in moderately noisy environments. The system was tested for continuous operation and ran for approximately 10 hours on a single 2000mAh battery charge. The wearable enclosure was found to be comfortable for extended wear during initial user trials.

The results validate that the system meets its primary functional requirements. The integration of multiple sensors and directional voice feedback provides a significant improvement over simpler systems. The user feedback from initial trials with sighted individuals simulating navigation with blindfolds was positive, with participants reporting that the directional cues were intuitive and greatly enhanced their confidence in navigating an obstacle course. The system successfully provides a "smart" layer of perception, fulfilling its aim to enhance mobility and safety for the visually impaired.

VI. CONCLUSION & REFERENCE

A.Conclusions

The VIORA smart assistive device successfully demonstrates the feasibility of using low-cost, IoT-enabled components to create a practical solution for the navigation challenges faced by the visually impaired. The system's strength lies in its simplicity and targeted functionality: providing clear, real-time, directional voice feedback about obstacles. By integrating three ultrasonic sensors with an ESP32 microcontroller and a voice feedback module, the device effectively translates physical environmental data into an intuitive auditory map for the user. The lightweight, wearable design ensures it can be incorporated into daily life without being a burden. This project successfully bridges the gap between complex, expensive electronic travel aids and the simple, yet limited, traditional white cane. It empowers users by providing them with the information they need to make informed decisions about their path, thereby increasing their safety, confidence, and independence. The ultimate outcome is a robust, affordable, and user-centered device that contributes to a more inclusive environment where mobility is not a barrier for the visually impaired.

B.Reference

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